Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The compressibility of water is 4×10^{-5} per unit atmospheric pressure. The decrease in volume of $100\ \mathrm{cm}^3$ of water under a pressure of 100 atmosphere will be

(A) $0.4\ \mathrm{cm}^3$

(B) $4 \times 10^{-5}\ \mathrm{cm}^3$

(C) $0.025\ \mathrm{cm}^3$

(D) $0.004\ \mathrm{cm}^3$

Q.2

One drop of soap bubble of radius D breaks into 27 drops having surface tension σ. The change in surface energy is

(A) $2\pi\sigma\mathrm{D}^2$

(B) $4\pi\mathrm{cD}^2$

(C) $\pi\sigma D^2$

(D) $8\pi\sigma D^2$

Q.3

When a capillary tube of radius r is immersed in a liquid of density ρ, the liquid rises to a height h in it. If m is the mass of the liquid in the capillary tube, the potential energy of this mass of the liquid in the tube is

(A) $mgh/4$

(B) $\mathrm{mgh}\,2$

(C) mgh

(D) $2mgh$

Q.4

A $20\ \mathrm{cm}$ long capillary tube is dipped in water. The water rises up to $8\mathrm{cm}$. If the entire arrangement put in a freely falling elevator the length of water column in the capillary tube will be

(A) $8\ \mathrm{cm}$

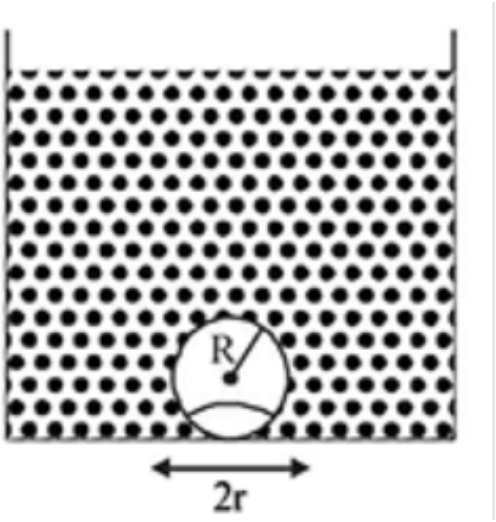
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(B) 10 cm

(C) 4 cm

(D) 20 cm

Q.5



On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel If $\ll R$ and the surface tension of water is T, value of r just before bubbles detach is: (density of water is ρ_w)

(A) $\mathrm{R}^2\sqrt{\frac{2\rho_{\mathrm{w}}\mathrm{g}}{3\,\mathrm{T}}}$

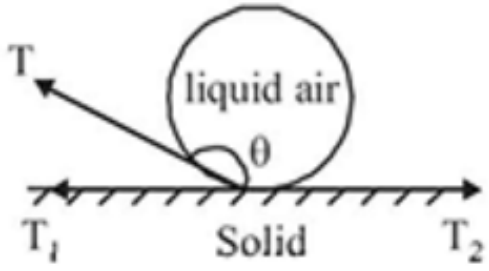
(B) $\mathrm{R}^2\sqrt{\frac{\rho_{\mathrm{w}}\mathrm{g}}{6\,\mathrm{T}}}$

(C) $\mathrm{R}^2\sqrt{\frac{\rho_{\mathrm{w}}\mathrm{g}}{\mathrm{T}}}$

(D) $\mathrm{R}^2\sqrt{\frac{3\rho_{\mathrm{w}}\mathrm{g}}{\mathrm{T}}}$

Q.6

Let T_1 be surface tension between solid and air, T_2 be the surface tension between solid and liquid and T be the surface tension between liquid and air. Then in equilibrium, for a drop of liquid on a clean glass plate, the correct relation is (θ is angle of contact)



(A) $\cos\theta = \frac{T}{T_1+T_2}$

(B) $\cos\theta = \frac{T}{T_1-T_2}$

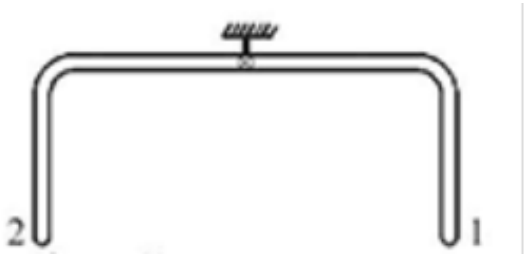
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(C) $\cos\theta = \frac{T_1+T_2}{T}$

(D) $\cos\theta = \frac{T_1-T_2}{T}$

Q.7

A glass tube of uniform internal radius
(r) has a valve separating the two identical ends. Initially, the valve is in
a tightly closed position. End 1 has a hemispherical soap bubble of radius r. End 2 has sub-hemispherical soap bubble (figure). Just after opening the valve



(A) air from end 1 flows towards end 2 . No change in the volume of the soap bubbles

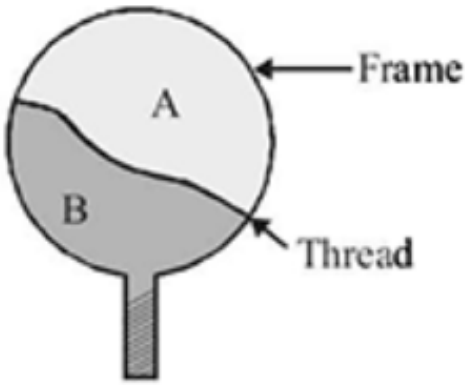
(B) air from end 1 flows towards end 2 . Volume of the soap bubble at end 1 decreases

(C) no change occurs

(D) air from end 2 flows towards end 1 . Volume of the soap bubble at end 1 increases

Q.8

A thread is tied slightly loose to a wire frame as in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is punctured with a pin, the thread



(A) becomes concave towards A

(B) becomes convex towards A

(C) remains in the initial position

(D) either (a) or (b) depending on the size of A w.r. t. B

Q.9

Wax is coated on the inner wall of a capillary tube and the tube is then dipped in water. Then, compared to the unwaxed capillary, the angle of contact θ and the height h upto which water rises change. These changes are :

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(A) θ increases and h also increases

(B) θ decreases and h also decreases

(C) θ increases and h decreases

(D) θ decreases and h increases

Q.10

Two mercury drops (each of radius ' r') merge to form bigger drop. The surface energy of the bigger drop, if T is the surface tension, is :

(A) $4\pi r^2 T$

(B) $2\pi r^2 T$

(C) $2^{8/3}\pi r^2 T$

(D) $2^{5/3}\pi r^2 T$

Q.11

A capillary tube of radius r is immersed vertically in a liquid such that liquid rises in it to height h (less than the length of the tube). Mass of liquid in the capillary tube is m. If radius of the capillary tube is increased by 50%, then mass of liquid that will rise in the tube, is

(A) $\frac{2}{3}$ m

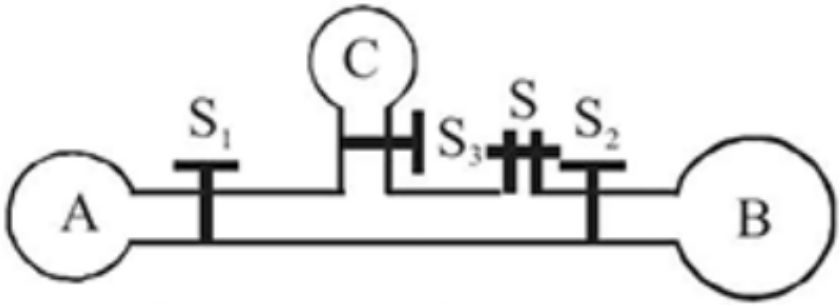
(B) m

(C) $\frac{3}{2}$ m

(D) $\frac{9}{4}$ m

Q.12

The diagram below shows three soap bubbles A, B and C prepared by blowing the capillary tube fitted with stop cocks $\mathrm{S}, \mathrm{S}_1, \mathrm{S}_2$ and S_3. When all the three stop cocks $\mathrm{S}_1, \mathrm{S}_2$ and S_3 are opened and S closed, then



(A) B will start collapsing with the increasing volume of A and C

(B) C will start collapsing with the increasing volume of A and B

(C) volumes of A, B and C will become equal at equilibrium

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(D) C and A will both start collapsing with the increasing volume of B

Q.13

An isolated and charged spherical soap bubble has a radius r and the pressure inside is atmospheric. T is the surface
tension of soap solution. If charge on drop is $\mathrm{X}\pi\mathrm{r}\sqrt{2\mathrm{r}\mathrm{T}\varepsilon_0}$
then find the value of X.

(A) 7

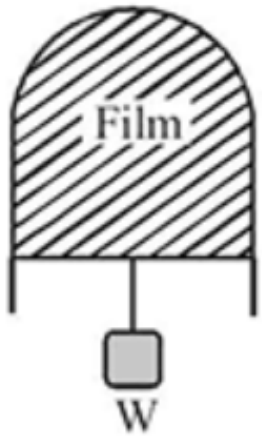
(B) 6

(C) 9

(D) 8

Q.14

A thin liquid film formed between a U-shaped wire and a light slider supports a weight of $1.5 \times 10^{-2}\ \mathrm{N}$ (see figure). The length of the slider is $30\ \mathrm{cm}$ and its weight negligible. What will be the surface tension (in Nm^{-1}) of the liquid film?



(A) 0.052

(B) 0.250

(C) 0.520

(D) 0.025

Q.15

If T_{SA} bet the surface tension between solid and air and T_{LA}, the surface tension between liquid and air and T st the surface tension between solid and liquid then the angle of contact θ of the liquid with the solid is

(A) $\cos^{-1}\left\{\frac{\mathrm{T}_{\mathrm{SA}}-\mathrm{T}_{\mathrm{SL}}}{\mathrm{T}_{\mathrm{LA}}}\right\}$

(B) $\tan^{-1}\left\{\frac{\mathrm{T}_{\mathrm{SA}}-\mathrm{T}_{\mathrm{SL}}}{\mathrm{T}_{\mathrm{X}}\mathrm{A}}\right\}$

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(C) $\cos^{-1}\left\{\frac{T_{SA}+T_L}{T_{SL}}\right\}$

(D) $\tan^{-1}\left\{\frac{T_{2A}+T_{IA}}{T_{SL}}\right\}$

Q.16

A number of identical water drops merge with one another to form a bigger drop of water. In this process, which of the following statement is wrong?

(A) Surface area of the bigger drop is less than the total surface area of all the smaller drops

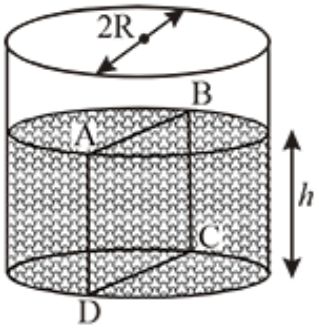
(B) Energy is released

(C) Temperature of the bigger drop is the same as that of a smaller drop

(D) Sum of the masses of all the smaller drops is equal to that of the bigger drop

Q.17

Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section $ABCD$ of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude



(A) $|\,2P_0Rh+\pi R^2\rho gh-2RT\,|$

(B) $|\,2P_0Rh+R\rho gh^2-2RT\,|$

(C) $|\,P_0\pi R^2+R\rho gh^2-2RT\,|$

(D) $|\,P_0\pi \mathbb{R}^2+R\rho gh^2+2RT\,|$

Q.18

A small hollow sphere, which has a small hole in it, is immersed in water to a depth for $0.5\ m$ before any drop penetrates into it. If surface tension for water is $0.073\ N/m$, the radus of the hole is

(A) $0.03\ mm$

(B) $0.08\ mm$

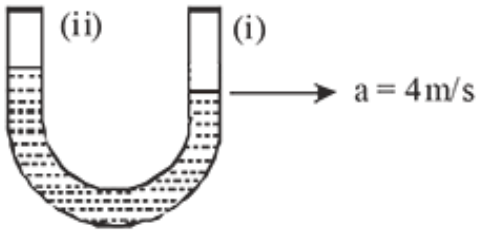
(C) $0.09\ mm$

(D) $0.15\ mm$

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Q.19

STATEMENT - 1: lo the given figure the liquid level in (it) is more than in (i).
STATEMENT - 2: The surface tension in tube (i) is lesser than in tube (ii).



(A) Statement -1 is True, Statement -2 is True; Statement - 2 is a correct explanation for Statement 1

(B) Statement - 1 is True, Statement - 2 is True; Statement - 2 is NOT a correct explanation for Statement -1

(C) Statement - 1 is True, Statement -2 is False.

(D) Statement -1 is False, Statement -2 is True

Q.20

A needle of length l and density ρ will float on a liquid of surface tension σ if its radius r is less than or equal to

(A) $\sqrt{\frac{2\sigma}{\pi\rho lg}}$

(B) $\sqrt{\frac{2\alpha}{\pi\rho g}}$

(C) $\sqrt{\frac{\sigma}{\pi pg}}$

(D) $\sqrt{\frac{2\sigma}{\pi\rho g}}$

Answers & Solutions

Q.1 Answer:

$0.4\ \text{cm}^3$

Solution:

Compressibility $(\beta) = \frac{\frac{\Delta V}{V}}{P}$ **Here,** $\beta = 4 \times 10^{-5}/$ **atm**

$\beta = 4 \times 10^{-5}/\text{atm}$

$\text{V} = 100\text{cc}$

$\text{P} = 100\ \text{atm}$

$\therefore \Delta\text{V} = \beta\text{PV}$

$= 4 \times 10^{-5} \times 100 \times 100$

$\therefore \triangle\text{V} = 0.4\text{cc}$

Q.2 Answer:

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$8\pi\sigma D^2$

Solution:

Volume of bigger bubble = volume of 27 smaller bubbles

$\Rightarrow \frac{4}{3}\pi D^3 = 27 \times \frac{4}{3}\pi d^3 \Rightarrow d = \frac{D}{3}$

Initial surface energy $\mathrm{S_i} = 4\pi \mathrm{D}^2\boldsymbol{\sigma}$

Final surface energy $S_t = 27 \times 4\pi d^2\sigma$ $\Delta \mathrm{S} = \mathrm{S_t} - S_\mathrm{i}$ **and using** $\mathrm{d} = \frac{\mathrm{D}}{3}$

$\Delta \mathrm{S} = \sigma \times 4\pi \left[27 \times \frac{\mathrm{D}^2}{9} - \mathrm{D}^2\right]$

$= 2\mathrm{D}^2 \times 4\pi \times \sigma = 8\pi\sigma \mathrm{D}^2$

Q.3 Answer:

$\mathrm{mgh}\,2$

Solution:

The mass of the liquid in a column between x **and** $x + dx$ **is**

$dx = \pi r^2 \rho dx$

Therefore, the potential energy of the liquid in a column of height h **is**

$\int_0^h \left(\pi r^2\right) x\rho g dx = \pi^2 \rho g \int_0^h x dx$

$= \pi^2\,\mathrm{sg}\,\frac{h^2}{2}$

$= \left(\pi r^2 hp\right) gh2 = mgh2$

Q.4 Answer:

$20\ \mathrm{cm}$

Solution:

As the system becomes weightless.

Therefore, liquid will rise to full height.

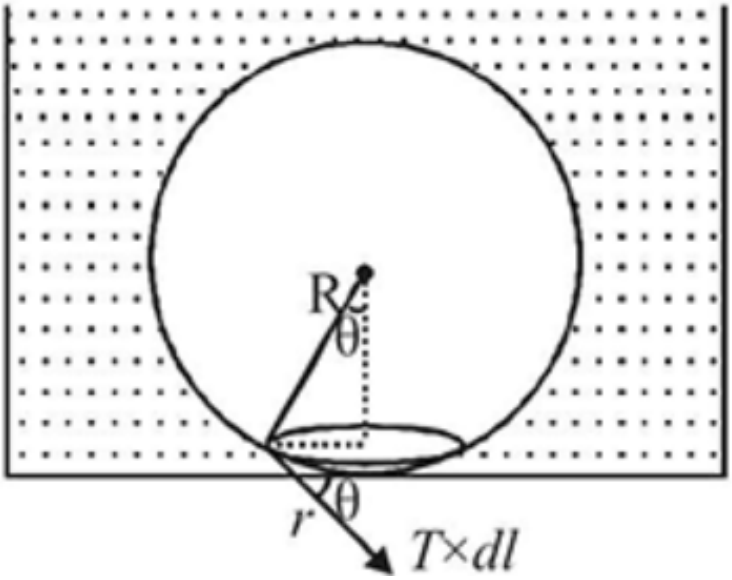
Q.5 Answer:

$\mathrm{R}^2\sqrt{\frac{2\rho_\mathrm{w}\mathrm{g}}{3\,\mathrm{T}}}$

Solution:

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When the bubble gets detached, Buoyant force = force due to surface tension



Force due to excess pressure = upthrust Access pressure in air bubble $= \frac{2\,\mathrm{T}}{\mathrm{R}}\ \frac{2T}{R}\left(\pi r^2\right) = \frac{4\pi R^3}{3}\rho_w g$

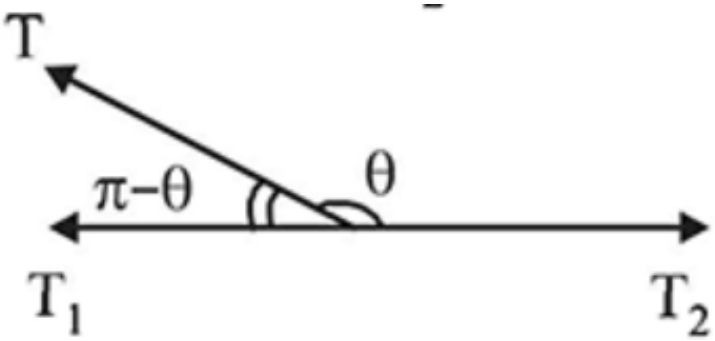
$\Rightarrow \quad r^2 = \frac{2R^4\rho_w g}{3T} \Rightarrow r = R^2\sqrt{\frac{2\rho_w g}{3T}}$

Q.6 Answer:

$\cos\theta = \frac{T_1 - T_2}{T}$

Solution:

$\therefore \quad \cos(\pi - \theta) = \frac{T_2 - T_1}{T}$



$\therefore \quad -\cos\theta = \frac{T_2 - T_1}{T}$

$\therefore \quad \cos\theta = \frac{T_1 - T_2}{T}$

Q.7 Answer:

air from end 1 flows towards end 2 . Volume of the soap bubble at end 1 decreases

Solution:

Pressure inside tube $= P = P_0 + \frac{4T}{r}$

$\therefore P_2 < P_1$ (**since** $r_2 > r_1$)

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Hence pressure on side 1 will be greater than side 2. So air from end 1 flows towards end 2

Subhemispherical radius $= \mathrm{r}_2$ **Hemispherical radius** $= \mathrm{r}_2$ **radius** $= \mathrm{r}_1$

Q.8 Answer:

becomes concave towards A

Solution:

Because film tries to cover minimum surface area.

Q.9 Answer:

θ increases and h decreases

Solution:

Angle of contact θ

$$\cos\theta = \frac{\mathrm{T_{SA}} - \mathrm{T_{SL}}}{\mathrm{T_{LA}}}$$

when water is on a waxy or oily surface $\mathrm{T_{SA}} < \mathrm{T_{SL}}$ $\cos\theta$ **is negative i.e.,** $90^\circ < \theta < 180^\circ$

i.e., angle of contact θ **increases And for** $\theta > 90^\circ$ **liquid level in capillary tube fall. i.e.,** h **decreases**

Q.10 Answer:

$2^{8/3}\pi r^2 T$

Solution:

Sum of volumes of 2 smaller drops = Volume of the bigger drop

$2 \cdot \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow R = 2^{1/3} r$

Surface energy $= T.4\pi R^2 = T4\pi 2^{2/3} r^2 = T.2^{8/3}\pi r^2$

Q.11 Answer:

$\frac{3}{2}$ m

Solution:

$$\mathrm{h} = \frac{2\,\mathrm{T}\cos\theta}{\mathrm{rog}} \Rightarrow \mathrm{h} \propto \frac{1}{\mathrm{r}} \Rightarrow \frac{\mathrm{h_2}}{\mathrm{h_1}} = \frac{\mathrm{r_1}}{\mathrm{r_2}} = \frac{2}{3}$$

$$\left(\because \mathrm{r_1} = \mathrm{r}, \quad \mathrm{r_2} = \mathrm{r} + 50\%\ \text{of}\ \mathrm{r} = \frac{3}{2}\mathrm{r}\right)$$

$$\text{New mass}\ \mathrm{m_2} = \pi \mathrm{r_2^2}\, \mathrm{h_2}\rho = \pi \left(\frac{3}{2}\mathrm{r_1}\right)^2 \left(\frac{2}{3}\,\mathrm{h_1}\right)\rho$$

$$= \frac{3}{2}\left(\pi \mathrm{r_1^2}\,\mathrm{h_1}\right)\rho = \frac{3}{2}\,\mathrm{m}$$

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Q.12 Answer:

C and A will both start collapsing with the increasing volume of B

Solution:

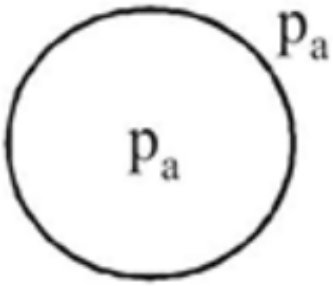
The excess pressure inside a bubble of soap is given by $p = \frac{4T}{r}$. It means excess pressure inside the bubbles A and C is more than inside B. So, air will go towards B from A and C. So, A and C will start collapsing with the increasing volume of B.

Q.13 Answer:

8

Solution:

Inside pressure must be $\frac{4T}{r}$ greater than outside pressure in bubble.



This excess pressure is provided by charge on bubble.

$\frac{4T}{r} = \frac{\sigma^2}{2\varepsilon_0}; \quad \frac{4T}{r} = \frac{Q^2}{16\pi^2 r^4 \times 2\varepsilon_0} \left[\sigma = \frac{Q}{4\pi r^2}\right]$

$Q = 8\pi r\sqrt{2rT\varepsilon_0}$

Q.14 Answer:

0.025

Solution:

At equilibrium, weight of the given block is balanced by force due to surface tension, i.e.,

$2\,L.S = W$

or $S = \frac{W}{2L} = \frac{1.5\times10^{-2}\ N}{2\times0.3\ m} = 0.025 Nm^{-1}$

Q.15 Answer:

$\cos^{-1}\left\{\frac{T_{SA}-T_{SL}}{T_{LA}}\right\}$

Solution:

$$\cos\theta = \frac{T_{SA}-T_{SL}}{T_{LA}} \Rightarrow \boldsymbol{\theta} = \cos^{-1}\left\{\frac{T_{SA}-T_{SL}}{T_{LA}}\right\}$$

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Q.16 Answer:

Temperature of the bigger drop is the same as that of a smaller drop

Solution:

When n identical liquid drop unit to form a bigger drop.

(i) surface area of the bigger drop decreases

(ii) volume (or mass) is the same as that of n smaller drops

(iii)energy is released duriog the process (iv)temperature increases Similarly, when a bigger drop breaks up into n smaller identical drops

(i) surface area of the drops increases

(ii) volume (or mass) of all the 'n' drops remains the same as that of the bigger drop

(iii)energy is absorbed

(iv) temperature decreases.

Q.17 Answer:

$| \mathbf{2P_0Rh} + \mathrm{R}\rho\mathrm{gh}^2 - \mathbf{2RT} |$

Solution:

$\mathrm{R} =$| **(pressure at the centre) (area of section ABCD) + (surface tension) (length AB)** $\mathbf{F} = \left|\left(\mathbf{P}_0 + \frac{\mathbf{h}}{2}\boldsymbol{Pg}\right)(\mathbf{2Rh}) - \mathbf{T}2\mathbf{R}\right|$
$=| \mathbf{2P_0Rh} + \mathbf{R}\,\mathrm{pgh}^2 - \mathbf{2RT}$

Q.18 Answer:

$0.03\ \mathrm{mm}$

Solution:

Since liquid drop will penetrate if pressure on both side will be same

$\therefore \quad P_0 + h\ell g = \rho_0 + 2T/R$

$\therefore \mathrm{heg} = \frac{2\,\mathrm{T}}{\mathrm{R}}$

$\therefore \mathrm{R} = \frac{2\,\mathrm{T}}{\mathrm{b}\ell\mathrm{g}} = \frac{2\times0.073}{0.5\times1000\times10} = .03\ \mathrm{mm}$

Q.19 Answer:

Statement - 1 is True, Statement -2 is False.

Solution:

No solution available

Q.20 Answer:

$\sqrt{\frac{2\sigma}{\pi\rho g}}$

Solution:

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Rorce per unit area exerted on the needle due to liquid is

$\mathbf{F_1 = \frac{\sigma}{r}}$

Pressure exerted by needle on the liquid is \mathbf{P}_{2}=\frac{\mathrm{mg}}{l(2 \mathrm{r})}=\frac{\pi \mathrm{d}^{2} \log }{2 \boldsymbol{l}}=\frac{\pi \mathrm{x} \mathrm{\rhog}}{2}

For equilibrium, $\mathrm{F_1 = P_2}$**, i.e.**

$\frac{\sigma}{2} = \frac{\pi x p g}{2}$

or $x = \sqrt{\frac{2\sigma}{\pi \rho g}}$

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